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Grounding Time: Temporally-Aware VideoQA System

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Abstract

Temporally grounded Video Question Answering requires a model not only to answer a question correctly but also to identify the temporal evidence that supports its prediction. This setting is particularly challenging for multiple-choice video reasoning, where raw answer candidates are often suboptimal for temporal localization, and correct prediction depends on both reliable evidence retrieval and grounded reasoning over the retrieved moments. In this paper, we present a modular framework for temporally grounded VideoQA consisting of four components: a *Refiner*, a *Retriever*, a *Validator*, and a *Generator*. The Refiner first rewrites answer candidates into grounding-friendly queries, enabling the Retriever to localize more relevant temporal evidence. The retrieved candidate moments are then scored and ranked by the Validator, and the final answer is produced by the Generator based on the highest-ranked evidence rather than the full video. To support this design, we adopt a role-specific strategy in which the Retriever and Validator are optimized for their designated functions, while the Generator serves as the final grounded reasoning module. Experiments on NExT-GQA and ReXTime show that our framework improves both temporal grounding quality and grounded question answering accuracy, with especially clear gains under strict grounding-aware evaluation. These results highlight the importance of combining answer-aware query refinement, stronger temporal retrieval, explicit evidence validation, and grounded answer generation for reliable video understanding.

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1. Introduction

Understanding videos requires not only recognizing visual content but also identifying when relevant events occur. This requirement is especially critical in temporally grounded Video Question Answering (VideoQA), where a model must both predict the correct answer and localize the temporal evidence that justifies it. In this setting, answer correctness alone is insufficient: a prediction is reliable only if it is supported by faithful evidence rather than language priors or spurious visual correlations [1, 2].

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Recent multimodal large language models (MLLMs) have made substantial progress in video understanding, including temporal grounding and long-video reasoning [3, 4]. Methods such as VTimeLLM, TimeChat, and Grounded-VideoLLM improve temporal sensitivity and grounding ability, while recent reasoning frameworks such as SeViLA, LLoVi, and VideoMind demonstrate that decomposing video reasoning into multiple stages can enhance performance and interpretability [5–10]. Despite these advances, a fundamental challenge remains underexplored in *multiple-choice grounded VideoQA*.

Specifically, we observe a structural mismatch between *answer formulation* and *temporal evidence retrieval*. In multiple-choice VideoQA, answer candidates are designed for discrimination, often being abstract, incomplete, or weakly visual. However, temporal grounding requires queries that explicitly describe observable events. As a result, directly using answer candidates as retrieval queries leads to systematic errors: even strong temporal grounding models may retrieve segments that are temporally close but semantically weak or only partially aligned with the required evidence. This mismatch limits the effectiveness of existing approaches, which typically assume that answer representations can be directly grounded.

To address this problem, we propose a modular framework for temporally grounded VideoQA that explicitly separates *evidence reformulation*, *evidence acquisition*, *evidence validation*, and *answer generation*. Our framework consists of four components: a Refiner, a Retriever, a Validator, and a Generator. The Refiner first rewrites each answer candidate into a grounding-friendly query that exposes the underlying visual evidence. The Retriever then localizes candidate temporal spans for each refined query. Next, the Validator scores and ranks all retrieved candidates according to their reliability as supporting evidence. Finally, the Generator predicts the answer conditioned on the highest-ranked evidence rather than the entire video. This design enables each component to specialize in a distinct role, improving both evidence quality and grounded reasoning.

We evaluate our approach on NExT-GQA and ReXTime, two benchmarks that emphasize grounded answering and reasoning across time [1, 2]. Experimental results show that our method consistently improves performance on strict grounding-aware metrics, indicating more precise and reliable evidence localization. Moreover, we observe corresponding gains in grounded answer accuracy, demonstrating that better evidence directly translates into more reliable predictions.

Our contributions are summarized as follows:

- We identify a key mismatch in multiple-choice grounded VideoQA: answer candidates are suitable for discrimination but not for temporal evidence retrieval, leading to systematic retrieval errors.
- We propose a modular framework that resolves this mismatch through query refinement and explicit evidence verification, improving the alignment between retrieval and reasoning.
- We achieve strong performance on NExT-GQA and ReXTime, demonstrating that improving evidence quality leads to more accurate and reliable grounded VideoQA.

2. Related Work

2.1. Temporal Video Grounding

Video Temporal Grounding (VTG) aims to localize the temporal segment in a video that corresponds to a language query. Recent progress in VTG increasingly leverages multimodal large language models (MLLMs), shifting from task-specific localization pipelines toward more unified video-language frameworks [4]. Representative methods such as VTimeLLM [5], TimeChat [6], and LITA [11] improve timestamp sensitivity, temporal representation, and localization-oriented instruction tuning for video understanding. Grounded-VideoLLM further strengthens fine-grained localization through grounding-oriented architectural design and multi-stage training [7].

Beyond supervised grounding, training-free and post-training paradigms have also shown promising results. Training-free VTG methods demonstrate that large pre-trained models already contain useful temporal reasoning priors [12–14], while more recent approaches such as Time-R1 and TimeLens highlight the importance of grounding-specific post-training, stronger data curation, and more reliable optimization for precise temporal localization [15, 16]. Overall, these works substantially improve the ability of MLLMs to localize relevant temporal moments. However, they mainly study VTG in the conventional query-to-moment setting. In contrast, our work focuses on *multiple-*

choice grounded VideoQA, where the input answer candidates are not naturally written as retrieval-friendly grounding queries, creating an additional mismatch between answer formulation and temporal evidence acquisition.

2.2. Grounded and Reasoning VideoQA

Conventional VideoQA primarily evaluates answer correctness, often without requiring the model to identify the evidence supporting its prediction. This limitation has motivated grounded VideoQA, where correct answering must be accompanied by faithful temporal evidence [1]. NExT-GQA is a representative benchmark in this direction, showing that strong answer accuracy does not necessarily imply grounded reasoning [1]. ReXTime complements this setting by emphasizing reasoning across time, where the evidence needed for answering may appear in a different temporal region from the question context [2]. More broadly, recent empirical studies in the LLM era also suggest that answer accuracy alone is insufficient for assessing genuine video understanding [17].

Several recent works attempt to connect temporal localization and answer prediction more explicitly. SeViLA decomposes the task into localization and answering stages, reducing reliance on dense temporal annotations through self-refinement [8]. For long-video reasoning, LLoVi, LangRepo, and VideoStreaming show that staged language-centric processing, memory-like representations, and streaming mechanisms can improve reasoning over temporally extended videos [9, 18, 19]. More recently, VideoExpert and VideoMind demonstrate that stronger temporal grounding and role specialization can further benefit grounded reasoning, especially in long-form and temporally complex settings [10, 20].

Our work is most closely related to this line of grounded reasoning methods, but differs in its task-specific focus. Existing approaches typically connect localization and answering more directly, whereas we target a distinct bottleneck in *multiple-choice* grounded VideoQA: answer options are designed for discrimination rather than temporal evidence retrieval. To address this mismatch, we explicitly introduce a refinement stage to transform answer candidates into retrieval-friendly queries and a validation stage to score and rank retrieved evidence before final answer generation.

3. Method

3.1. Overview

Given a video V, a question q, and a set of answer candidates $O = \{o_i\}_{i=1}^{N}$, temporally grounded VideoQA aims to predict not only the correct answer $\hat{y}$ but also the temporal evidence $\hat{s}$ that supports the prediction [1, 2]. Our method addresses a key mismatch in multiple-choice grounded VideoQA: answer candidates are written for discrimination, while temporal retrieval requires evidence-oriented queries. To bridge this gap, we decompose the pipeline into four components: a *Refiner*, a *Retriever*, a *Validator*, and a *Generator*.

For each answer candidate, the Refiner first rewrites it into a grounding-friendly query. The Retriever then localizes top-K candidate temporal spans for each refined query. Next, the Validator evaluates all retrieved clips and assigns them validation scores. Finally, the Generator predicts the answer conditioned on the highest-ranked evidence rather than the entire video. This decomposition separates evidence acquisition from evidence validation and answer generation, allowing each component to focus on a dedicated role.

3.2. Refiner

A key challenge in multiple-choice grounded VideoQA is that raw answer candidates are often not ideal temporal retrieval queries. They are typically written for answer discrimination rather than evidence localization, and may be too abstract, incomplete, or weakly visual to retrieve the correct temporal segment. This issue is particularly severe in grounded VideoQA, where answer correctness depends on accurate evidence localization [1, 2].

To address this problem, we introduce a query refinement stage. Given a question q and an answer candidate o_i, we use Gemini 2.5 Flash to rewrite the candidate into a grounding-friendly query:

$$r_i = \mathcal{F}_{\text{ref}}(q, o_i). \tag{1}$$

The goal of this step is to preserve the semantics of the answer candidate while making the underlying visual evidence more explicit. In practice, the Refiner transforms answer candidates from decision-oriented text into retrieval-oriented text, enabling the downstream Retriever to focus on visually localizable events rather than abstract answer semantics.

This design is especially helpful for causal and reasoning-across-time questions. In such cases, an answer candidate may refer to an inferred event or a temporally displaced consequence rather than a directly observable moment. Query refinement helps expose the visual evidence that should be retrieved before final answer generation. The Refiner is used as an off-the-shelf query rewriting module and is not further trained in our framework. The Refiner is a modular component and can be replaced by other LLMs.

3.3. Retriever

Given a refined query r_i, the Retriever localizes the top-K candidate temporal spans

$$\{s_{i,k}\}_{k=1}^{K} = \mathcal{F}_{\text{ret}}(V, r_i), \tag{2}$$

where each candidate span is represented as

$$s_{i,k} = [t_{i,k}^{s}, t_{i,k}^{e}], \tag{3}$$

with $t_{i,k}^{s}$ and $t_{i,k}^{e}$ denoting the predicted start and end timestamps. The corresponding retrieved clips are denoted by

$$c_{i,k} = V[s_{i,k}], \qquad k = 1, \ldots, K. \tag{4}$$

In our framework, the Retriever is initialized from Qwen3-VL-8B and fine-tuned for video temporal grounding on the TimeLens-100K dataset [16]. Following TimeLens, temporal information is injected through *interleaved textual timestamp encoding*, where timestamps are inserted as text tokens alongside the visual sequence. This design improves alignment between language queries and temporally grounded visual evidence.

The Retriever is trained with a three-stage pipeline. First, a supervised fine-tuning (SFT) warm-start is performed on 30K sampled instances from TimeLens-100K to initialize temporal grounding ability. Second, the SFT checkpoint is used to run offline inference on the full TimeLens-100K set, and the predicted spans are filtered according to temporal overlap quality. Third, the model is further optimized with GRPO under a thinking-free RLVR paradigm, where the reward is defined based on temporal localization quality. This training strategy improves the precision and reliability of temporal span prediction.

In our setting, the Retriever is applied to the refined queries produced from the answer candidates. For each refined query r_i, it returns candidate temporal spans that may serve as supporting evidence for answering the question. This design allows retrieval to operate on evidence-oriented reformulations rather than raw answer options, thereby improving the compatibility between temporal grounding and multiple-choice grounded VideoQA.

3.4. Validator

The Retriever returns a set of candidate evidence clips, but not every retrieved candidate is equally reliable for grounded reasoning. In grounded VideoQA, a candidate segment may be temporally close to the target event while still being weakly informative, noisy, or only partially aligned with the evidence required to answer the question. To improve the reliability of downstream reasoning, we introduce a *Validator* that verifies the retrieved candidates and assigns each one a score indicating how likely it is to serve as faithful evidence.

We denote the pool of candidate clips returned by the Retriever as

$$C = \{c_{i,k} \mid i = 1, \ldots, N;\ k = 1, \ldots, K\}, \tag{5}$$

where $c_{i,k}$ denotes the k-th candidate clip associated with the i-th query. Given a question q, the Validator assigns each candidate clip a validation score

$$\alpha_{i,k} = \mathcal{F}_{\text{val}}(q, c_{i,k}), \tag{6}$$

which measures how likely the candidate is to constitute faithful evidence for answering q. These scores induce a ranking over all retrieved candidates, which is then used by the Generator.

Our Validator follows the **verifier** design of VideoMind [10]. For each candidate clip, we adopt a *zoom-in* strategy by expanding its temporal boundaries by 50% on both sides and cropping the enlarged segment as input. Two special tokens, `<SEG-START>` and `<SEG-END>`, are inserted at the temporal positions corresponding to the original candidate boundaries, allowing the model to distinguish the hypothesized evidence span from its surrounding temporal context [10].

We formulate validation as a binary verification task. During training, each candidate is assigned a binary label according to its temporal overlap with the annotated evidence span: candidates with Intersection-over-Union (IoU) greater than 0.5 are treated as positive, and the others as negative, following [10]. The Validator is then fine-tuned to predict whether the highlighted segment should be accepted (`Yes`) or rejected (`No`) as valid evidence for the question.

During inference, for each candidate clip, we compute the logits of the `<Yes>` and `<No>` tokens under teacher forcing, denoted by L_y and L_n, respectively. The validation score is defined as

$$\alpha_{i,k} = \sigma(L_y - L_n), \tag{7}$$

where $\sigma(\cdot)$ is the sigmoid function. The resulting scores are used to rank all candidates before generation.

This validation stage is particularly important in grounded VideoQA, where answer prediction should rely on faithful evidence rather than merely plausible temporal proximity. By explicitly validating retrieved candidates before downstream reasoning, the Validator helps reduce the effect of near-miss retrievals and improves the robustness of grounded prediction [1, 10].

However, validation alone does not uniquely determine the correct answer. In practice, multiple answer candidates may correspond to visually similar or temporally overlapping events, making the evidence–answer alignment ambiguous. The highest-ranked clip therefore serves as grounded evidence for the subsequent Generator, which performs the final semantic reasoning over the candidate answers.

3.5. Generator

After validation, the Generator takes the highest-ranked evidence clip

$$\tilde{c} = c_{i^*,k^*}, \qquad (i^*, k^*) = \arg\max_{i,k} \alpha_{i,k}, \tag{8}$$

and predicts the final answer conditioned on the question, the answer candidates, and this selected evidence:

$$\hat{y} = \mathcal{F}_{\text{gen}}(q, O, \tilde{c}). \tag{9}$$

While the Validator ranks the retrieved candidates by reliability, evidence ranking alone is not sufficient for final answer prediction. In grounded multiple-choice VideoQA, multiple answer candidates may correspond to visually similar or temporally overlapping events, so the highest-ranked evidence does not always map uniquely to a single answer option. The role of the Generator is therefore to resolve this ambiguity by performing semantic reasoning over the selected evidence together with the question and the candidate answers.

In our implementation, the Generator is based on **Qwen2-VL** [21] and is used without additional task-specific training. Unlike the Retriever and Validator, which are explicitly optimized for evidence acquisition and evidence validation, respectively, the Generator serves as the final multimodal reasoning module. Conditioning the Generator on the highest-ranked evidence rather than the full video encourages grounded prediction and reduces distraction from irrelevant visual context, which is particularly important for long-video reasoning and grounded VideoQA [1, 10].

4. Experiments

4.1. Experimental Setup

Benchmarks. We evaluate our framework on two temporally grounded VideoQA benchmarks, **NExT-GQA** [1] and **ReXTime** [2]. NExT-GQA emphasizes visually grounded answering, requiring the model to predict the correct answer while localizing the supporting temporal evidence [1]. ReXTime focuses on reasoning across time, where the evidence required for answering may lie in a temporal region different from that implied by the question context [2].

Evaluation metrics. For NExT-GQA, following prior work [1, 10], we report both IoP-based and IoU-based grounding metrics, including mIoP, IoP@0.3, IoP@0.5, mIoU, IoU@0.3, and IoU@0.5, together with Acc@GQA (IoP>0.5) and Acc@QA. For ReXTime, following the benchmark protocol [2], we report R@0.3, R@0.5, mIoU, Acc, and Acc@IoU=0.5.

Baselines. On NExT-GQA, we compare against representative grounded or temporally aware VideoQA methods, including FrozenBiLMNG+ [22], VTimeLLM [5], SeViLA [8], LangRepo [18], VideoStreaming [19], Grounded-VideoLLM [7], VideoExpert [20], VideoChat-TPO [23], LLoVi [9], HawkEye [24], DeVi [25], and VideoMind [10]. On ReXTime, we compare against VTimeLLM [5], TimeChat [6], LITA [11], and VideoMind [10]. Unless otherwise noted, all baseline results are taken from the corresponding original papers.

Implementation details. Our framework consists of four modules: a Refiner, a Retriever, a Validator, and a Generator. The Refiner uses Gemini 2.5 Flash and is applied without additional training. The Retriever is initialized from Qwen3-VL-8B and fine-tuned on TimeLens-100K with a three-stage training pipeline consisting of SFT warm-start, offline filtering, and GRPO-based optimization. The Validator is built on Qwen2-VL-7B and trained with LoRA-based supervised fine-tuning following the verifier design of VideoMind [10]. The Generator is implemented with Qwen2-VL [21] and is used without additional fine-tuning. In all experiments, we set $K = 2$, where the Retriever generates two candidate temporal spans for each refined query, the Validator scores and ranks these candidates, and the Generator receives the highest-ranked clip for downstream reasoning. Table 1 summarizes the training configuration of the trainable modules.

Table 1. Training configuration of the trainable modules in our framework.

Module	Backbone	Training data	Training strategy
Retriever	Qwen3-VL-8B	TimeLens-100K	SFT → Filter → GRPO
Validator	Qwen2-VL-7B	DiDeMo, TACoS, QVHighlights	LoRA-based SFT

4.2. Main Results

As shown in Table 2, our method achieves the best overall performance among the compared methods on NExT-GQA, with particularly clear gains on strict grounding-aware metrics. Specifically, compared with the strongest prior baseline, VideoMind [10], our method improves mIoU by 3.46%, IoU@0.5 by 5.77%, and Acc@GQA by 2.85%, while also achieving the best mIoP and IoP@0.5.

A notable pattern is that our method performs slightly below VideoMind on the looser thresholds IoP@0.3 and IoU@0.3, but consistently outperforms it on the stricter thresholds IoP@0.5 and IoU@0.5. This suggests that the

Table 2. Results on NExT-GQA. PT denotes trained/pretrained models, and TF denotes training-free models. Best results are in bold.

Method	Paradigm	mIoP	IoP@0.3	IoP@0.5	mIoU	IoU@0.3	IoU@0.5	Acc@GQA	Acc@QA
FrozenBiLMNG+ [22]	PT	24.2	28.5	23.7	9.6	13.5	6.1	17.5	70.8
VTimeLLM [5]	PT	27.9	30.3	23.8	18.3	27.7	14.1	12.7	–
SeViLA [8]	PT	29.5	34.7	22.9	21.7	29.2	13.8	16.6	68.1
LangRepo [18]	PT	31.3	–	28.7	18.5	–	12.2	17.1	–
VideoStreaming [19]	–	32.2	–	31.0	19.3	–	13.3	17.8	–
Grounded-VideoLLM [7]	PT	34.5	42.6	34.4	21.1	30.2	18.0	26.7	–
VideoExpert [20]	PT	34.6	45.3	29.3	27.9	41.0	22.4	21.6	71.1
VideoChat-TPO [23]	PT	35.6	47.5	32.8	27.7	41.2	23.4	25.5	–
LLoVi [9]	PT	37.3	–	36.9	20.0	–	15.3	24.3	–
HawkEye [24]	PT	–	–	–	25.7	37.0	19.5	–	–
DeVi [25]	TF	39.7	–	38.9	23.6	–	19.5	28.9	73.1
VideoMind [10]	PT	39.0	**56.0**	35.3	31.4	**50.2**	25.8	28.2	76.9
Ours	PT	**41.76**	54.13	**39.02**	**34.86**	49.88	**31.57**	**31.05**	**77.0**

Table 3. Results on ReXTime. FT indicates the fine-tuned setting reported by prior work. Best results are in bold.

Method	Size	FT	R@0.3	R@0.5	mIoU	Acc	Acc@IoU=0.5
VTimeLLM [5]	7B	✓	43.69	26.13	29.92	57.58	17.13
TimeChat [6]	7B	✓	40.13	21.42	26.29	49.46	10.92
VTimeLLM [5]	7B	–	28.84	17.41	20.14	36.16	–
TimeChat [6]	7B	–	14.42	7.61	11.65	40.04	–
LITA [11]	13B	–	29.49	16.29	21.49	34.44	–
VideoMind [10]	7B	–	38.22	25.52	27.61	**74.59**	20.20
Ours	8B	–	**46.91**	**35.94**	**36.99**	**74.59**	**27.25**

proposed framework improves precise temporal localization rather than only coarse overlap. Since NExT-GQA explicitly evaluates whether predicted answers are supported by correct temporal evidence [1], these results indicate better alignment between answer prediction and grounded temporal evidence.

As shown in Table 3, our method delivers the best grounding performance among the compared methods on ReXTime, achieving the highest R@0.3, R@0.5, and mIoU, while matching the best answer accuracy. In particular, compared with VideoMind [10], our method improves mIoU by 9.38% and Acc@IoU=0.5 by 7.05%.

This result is especially important on ReXTime, where answering correctly requires identifying temporally displaced evidence [2]. In this setting, answer accuracy alone is insufficient; the model must also retrieve the correct answer-supporting span. The gains on both retrieval metrics and Acc@IoU=0.5 therefore suggest that our framework is more effective at linking answer-relevant temporal evidence to final predictions. Taken together, the results on NExT-GQA and ReXTime show that our method is particularly effective under strict grounding-aware evaluation, where answer correctness and evidence quality must be satisfied jointly.

4.3. Ablation Study

We conduct ablations on ReXTime [2], which offers a particularly discriminative testbed due to its reasoning-across-time nature. Table 4 compares three variants: a Generator-only baseline, a Retriever–Validator–Generator pipeline without the Refiner, and the full model.

Table 4. Ablation results on ReXTime. Ref, Ret, Val, and Gen denote Refiner, Retriever, Validator, and Generator, respectively. The Generator-only variant does not produce temporal evidence spans, so grounding metrics are not reported.

Ref	Ret	Val	Gen	R@0.3	R@0.5	mIoU	Acc	Acc@IoU=0.5
–	–	–	✓	–	–	–	68.00	–
–	✓	✓	✓	33.01	25.19	27.17	73.24	19.22
✓	✓	✓	✓	46.91	35.94	36.99	74.59	27.25

As shown in Table 4, the ablation study yields two clear observations. First, introducing explicit evidence retrieval and selection already provides a strong improvement over direct generation, raising answer accuracy from 68.00 to 73.24. This indicates that ReXTime cannot be solved reliably by generation alone, since correct answers often depend on evidence located at temporally displaced moments.

Second, the Refiner brings an additional and consistent gain across all reported metrics. Relative to the Retriever–Validator–Generator variant, the full model improves R@0.3 by 13.90%, R@0.5 by 10.75%, mIoU by 9.82%, and Acc@IoU=0.5 by 8.03%, while also increasing Acc from 73.24% to 74.59%. These results suggest that the Refiner

improves the quality of the evidence supplied to downstream modules, resulting in more precise temporal grounding and stronger grounded question answering.

Overall, the ablation study confirms that the effectiveness of the full framework does not arise from generation alone, but from progressively improving evidence quality through the Refiner–Retriever–Validator–Generator pipeline.

5. Conclusion

In this paper, we propose a modular framework for temporally grounded VideoQA consisting of four components: a Refiner, a Retriever, a Validator, and a Generator. Our design is motivated by the observation that reliable grounded answer prediction depends not only on strong reasoning ability, but also on the quality of the temporal evidence provided to the final answer generator. To address this challenge, we first use the Refiner to rewrite answer candidates into grounding-friendly queries, then use the Retriever to localize candidate evidence spans, the Validator to score and rank the retrieved evidence, and the Generator to produce the final answer based on the highest-ranked clip.

Experimental results on NExT-GQA and ReXTime show that our framework improves both temporal grounding quality and grounded question answering accuracy. In particular, the gains on stricter grounding-aware metrics indicate that the proposed pipeline is effective at producing more precise and reliable evidence, rather than merely retrieving roughly relevant moments. Our analysis further suggests that answer-aware refinement, explicit evidence selection, and evidence-conditioned generation all play important roles in achieving robust reasoning across time.

Overall, our findings show that temporally grounded VideoQA benefits from explicitly separating evidence refinement, retrieval, validation, and generation. In future work, it would be interesting to extend the framework to multi-span reasoning, stronger query rewriting, and more challenging long-form video settings where evidence may be sparse, compositional, or temporally distant.

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